

Interacting Multipoles of Rank 3 and 4:

[illegible]

Simplify deltas:

[illegible]

[illegible]

[illegible]

Simplify R:

[illegible]

$$\begin{aligned} \xrightarrow{\text{cleaned up}} R_z^{-15} \cdot & -\frac{1}{1575} \cdot \\ & +420 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\ & +630 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\ & +210 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\ & +420 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\ & +630 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\ & +210 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\zeta\zeta} \\ & +420 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\ & +630 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\ & +105 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ & +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ & +105 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ & +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ & +105 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\zeta\zeta} \\ & +630 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \\ & +210 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ & +105 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ & +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\delta\delta} \\ & +105 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ & +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\delta\delta} \end{aligned} \quad (3b)$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-15} \cdot -\frac{1}{1575} \cdot \left[\begin{aligned}
& +41580 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{zzz\gamma} + 41580 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{zzz\beta} \\
& +41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} - 1890 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& -3780 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} - 5670 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\delta\delta} \\
& -1890 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\zeta\zeta} - 3780 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& -5670 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\delta\delta} - 11340 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{zz\beta\gamma} \\
& -1890 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} - 3780 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\
& -5670 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\delta\delta} - 11340 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{zz\alpha\gamma} \\
& -11340 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{zz\alpha\beta} - 3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{zzz\gamma} \\
& -3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{zzz\beta} - 3780 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{zzz\alpha} \\
& +210 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\zeta\zeta} \\
& +420 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\
& +630 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\
& +210 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\
& +420 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\
& +630 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\
& +210 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\zeta\zeta} \\
& +420 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\
& +630 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\
& +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& +420 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\
& +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\zeta\zeta} \\
& +420 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& +210 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\zeta\zeta} \\
& +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \\
& +420 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\epsilon\epsilon} \\
& +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\delta\delta} \\
& +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\delta\delta} \\
& +630 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\delta\delta}
\end{aligned} \right] \tag{3c}
\end{aligned}$$

Simplify Hexadecapoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega_{\alpha\beta\gamma}\Phi_{\delta\epsilon\zeta\eta} = R_z^{-15} \cdot -\frac{1}{1575} \cdot \left[\begin{aligned} &+41580 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{zzz\gamma} + 41580 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{zzz\beta} \\ &+41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} - 1890 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ &-3780 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} - 5670 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\delta\delta} \\ &-1890 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\zeta\zeta} - 3780 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ &-5670 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\delta\delta} - 11340 \cdot R_z^7 \cdot \Omega_{z\beta\beta} \cdot \Phi_{zz\beta\beta} \\ &-1890 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} - 3780 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ &-5670 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\delta\delta} - 11340 \cdot R_z^7 \cdot \Omega_{z\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ &-11340 \cdot R_z^7 \cdot \Omega_{z\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} - 3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{zzz\gamma} \\ &-3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{zzz\beta} - 3780 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{zzz\alpha} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\zeta\zeta} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\zeta\zeta} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\zeta\zeta} \\ &\quad +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\delta\delta} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\delta\delta} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\delta\delta} \end{aligned} \right] \quad (4a)$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-15} \cdot -\frac{1}{1575} \cdot \left[\begin{aligned}
& +41580 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{zzz\gamma} + 41580 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{zzz\beta} \\
& +41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} - 1890 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& -3780 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} - 5670 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\delta\delta} \\
& -1890 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\zeta\zeta} - 3780 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& -5670 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\delta\delta} - 1890 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} \\
& -3780 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} - 5670 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\delta\delta} \\
& -3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{zzz\gamma} - 3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{zzz\beta} \\
& -3780 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{zzz\alpha} + 210 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\
& \quad +630 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\
& \quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\
& \quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\zeta\zeta} \\
& +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \\
& +420 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\epsilon\epsilon} \\
& +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{z\gamma\delta\delta} \\
& +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\delta\delta} \\
& +630 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\delta\delta}
\end{aligned} \right] \tag{4b}
\end{aligned}$$

Exploit Traceless Conditions:

$$\begin{aligned}
& T_{\alpha\beta\gamma\delta\epsilon\zeta\eta} \Omega_{\alpha\beta\gamma} \Phi_{\delta\epsilon\zeta\eta} = R_z^{-15} \cdot -\frac{1}{1575} \cdot \left[\begin{aligned}
& +41580 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{zzz\gamma} + 41580 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{zzz\beta} \\
& +41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} - 1890 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& -3780 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} - 5670 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\delta\delta} \\
& -1890 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\zeta\zeta} - 3780 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& -5670 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\delta\delta} - 1890 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} \\
& -3780 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} - 5670 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\delta\delta} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\
& \quad +630 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\
& \quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\
& \quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\zeta\zeta} \\
& \quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\
& \quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\
& +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma}
\end{aligned} \right] \tag{5}
\end{aligned}$$

Reduce Indices:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega_{\alpha\beta\gamma}\Phi_{\delta\epsilon\zeta\eta} = R_z^{-15} \cdot -\frac{1}{1575} \cdot \left[\begin{aligned} &+41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} + 41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} \\ &+41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} - 1890 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ &-3780 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\beta\beta} - 5670 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ &-1890 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\alpha\alpha\beta} - 3780 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\alpha\alpha\beta} \\ &-5670 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\alpha\alpha\beta} - 1890 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ &-3780 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\beta\beta} - 5670 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\alpha\alpha\beta\gamma} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\alpha\alpha\beta\gamma} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\alpha\alpha\beta\gamma} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\beta\beta\gamma} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\beta\beta\gamma} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\beta\beta\gamma} \\ &\quad +210 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \\ &\quad +420 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \\ &\quad +630 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \\ &\quad +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \end{aligned} \right] \quad (6a)$$

$$\xrightarrow{\text{added up}} R_z^{-15} \cdot -\frac{1}{1575} \cdot \left[\begin{aligned} &+124740 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} - 22680 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ &-11340 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\alpha\alpha\beta} + 1260 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{\alpha\alpha\beta\gamma} \\ &\quad +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{\alpha\beta\beta\gamma} \\ &\quad +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \\ &\quad +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \end{aligned} \right] \quad (6b)$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega_{\alpha\beta\gamma}\Phi_{\delta\epsilon\zeta\eta} = R_z^{-15} \cdot -\frac{1}{1575} \cdot \begin{aligned} & +1200 \cdot R_z^7 \cdot \Omega_{yyz} \cdot \Phi_{yyzz} + 1200 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{xyyz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yyyzz} + 1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yzzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xxxz} + 1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xyyz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xzzz} + 1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{xxyz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yyyzz} + 1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yzzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{zzz} \cdot \Phi_{xxxz} + 1260 \cdot R_z^7 \cdot \Omega_{zzz} \cdot \Phi_{yyzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{zzz} \cdot \Phi_{zzzz} + 1260 \cdot R_z^7 \cdot \Omega_{xxz} \cdot \Phi_{xxxz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xxz} \cdot \Phi_{xxyy} + 1260 \cdot R_z^7 \cdot \Omega_{xxz} \cdot \Phi_{xxzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xyz} \cdot \Phi_{xxxz} + 1260 \cdot R_z^7 \cdot \Omega_{xyz} \cdot \Phi_{xyyy} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xyz} \cdot \Phi_{xyzz} + 1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xxxz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xyyz} + 1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xzzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xyz} \cdot \Phi_{xxyy} + 1260 \cdot R_z^7 \cdot \Omega_{xyz} \cdot \Phi_{xyyy} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xyz} \cdot \Phi_{yzzz} + 1260 \cdot R_z^7 \cdot \Omega_{yyz} \cdot \Phi_{xxyy} \\ & +1260 \cdot R_z^7 \cdot \Omega_{yyz} \cdot \Phi_{yyyzz} + 1260 \cdot R_z^7 \cdot \Omega_{yyz} \cdot \Phi_{yzzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{xxyz} + 1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yyyzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yzzz} + 1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xxxz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xyyz} + 1260 \cdot R_z^7 \cdot \Omega_{xzz} \cdot \Phi_{xzzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{xxyz} + 1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yyyzz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{yzz} \cdot \Phi_{yzzz} + 1260 \cdot R_z^7 \cdot \Omega_{zzz} \cdot \Phi_{xxxz} \\ & +1260 \cdot R_z^7 \cdot \Omega_{zzz} \cdot \Phi_{yyzz} + 1260 \cdot R_z^7 \cdot \Omega_{zzz} \cdot \Phi_{zzzz} \end{aligned} \quad (7a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-15} \cdot -\frac{1}{1575} \cdot [+0] \quad (7b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega_{\alpha\beta\gamma}\Phi_{\delta\epsilon\zeta\eta} = +0 \quad (8)$$